Lesson 13: A Hit That Reacts

Scratch

VIDEO GAME CREATION

This week: getting hit finally LOOKS like getting hit. A brief flash, then a small explosion, before the enemy disappears.

Open your own [YourName]Game project from Lesson 12 - don't start a new one.

Important: your score, lives, wave, and win/lose logic should all still work exactly as before. Today only changes how the enemy's removal LOOKS, not when your game-state actually changes.

Setup

1. Open your [YourName]Game project from Lesson 12.
2. Add the explosion_placeholder.png costume to a brand-new sprite named Explosion (Scratch will let you draw/import a new sprite from a file).
3. Add "set [color] effect to 0" to the Enemy's existing "when I receive [spawn enemies]" spawn script (or wherever each enemy resets), so a fresh enemy never starts already flashed.

Level 1 - Minimal (MVP)

On every hit, the enemy briefly flashes, and a small independent explosion effect appears and cleans itself up. Score, lives, wave, and win/lose logic are all unaffected.



Everything in the top box happens the instant the hit is detected. Only the enemy's own removal - the very last step - waits for the pause to finish.

4. On the Explosion sprite, hide it by default, and add a new script: "when I receive [explosion]" -> show -> go to x/y of Enemy (using the enemy's position at the moment of the broadcast) -> wait 0.2 seconds -> hide.
5. In the Enemy's touching-Bullet branch, right after the existing "start sound [sfx_hit]" line, add: broadcast [explosion] (this spawns the visual independently, with zero effect on anything else) and "set [color] effect to 100" - both run immediately, same as the sound.
6. Your existing "broadcast [enemy hit]" stays exactly where it already is, right after those two new lines - GameManager still hears about the hit, and still updates score/enemies_remaining/wave-check, at exactly the same instant as last week. Nothing about that broadcast changes.
7. Add "wait 0.1 seconds" AFTER broadcast [enemy hit], then move "delete this clone" to run after that wait - this is the only thing actually delayed: the enemy sticks around, visibly flashed, for a moment before it's actually removed.
8. Click the stage and play through a full round - each hit should flash the enemy briefly, and a small burst should appear at that spot for a moment.

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Enemy's touching-Bullet branch (revised):
if <touching Bullet?> then
  start sound [sfx_hit]
  broadcast [explosion]
  set [color] effect to 100
  broadcast [enemy hit]
  wait 0.1 seconds
  delete this clone
```

Level 2 - Complete (Alpha/Beta)

The flash and explosion play correctly on every single hit, with no leftover explosions, no enemies stuck mid-flash, and no change to score/lives/wave timing.

- Play a full round: every single hit shows a flash and an explosion, with no enemy getting visually "stuck" flashed.
- Confirm score, wave-clear, and win/lose all still happen at exactly the moments they did last week - broadcast [enemy hit] hasn't moved at all, only delete this clone is delayed.
- Confirm the broadcast-based Explosion sprite never blocks or delays anything else - broadcasts don't wait for their receivers to finish.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Audio/Visual: instead of one flat 0.1-second flash, alternate the color effect a couple of times (e.g. 100, then 0, then 100 again) with short waits between, for a stronger flicker.
- Visual: give the Explosion sprite 2-3 different costumes and cycle through them with "next costume" during its 0.2-second lifetime for a simple hand-animated burst instead of one static image.
- Structure: notice the Explosion sprite knows nothing about score, lives, or waves - it does exactly one job, triggered independently of the rest of the hit-handling logic.

If Something's Wrong

- Enemy vanishes instantly, no flash visible: confirm "wait 0.1 seconds" was actually added, and that "delete this clone" was moved to AFTER it, not before.
- Explosion never appears: confirm the Explosion sprite actually has a "when I receive [explosion]" script, and that its costume isn't accidentally still hidden.
- Score changes twice per hit, or win-check fires early: confirm "broadcast [enemy hit]" wasn't accidentally duplicated or moved when adding the new lines - it should still appear exactly once, in its original spot.

Want More?

Student Resource List > Scratch section, or scratch.mit.edu/ideas